Machine Learning

Lab Evaluation Report

A Review

on

Multimodal Negative Emotion Recognition using EEG Signals and Facial Analysis

Submitted by Group 14-

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A Review on Multimodal Negative Emotion Recognition using EEG Signals and Facial Analysis

Abstract

Emotion recognition is an essential area of research in artificial intelligence, particularly in domains such as healthcare, human-computer interaction, and mental health monitoring. Traditional approaches to emotion recognition rely on observable cues such as facial expressions and speech signals; however, these methods often fail to capture the true internal emotional state of an individual. Electroencephalography (EEG) signals provide direct insight into brain activity and therefore offer a more reliable representation of emotions. This paper presents a comprehensive review of multimodal emotion recognition systems that combine EEG signals with facial expression analysis, with a particular focus on negative emotions. The study addresses key research questions related to the importance of emotion recognition, types of emotion recognition methods, the valence-arousal model, machine learning and deep learning techniques, and the necessity of multimodal approaches. Furthermore, a hybrid CNN-LSTM-based model is proposed, and a comparative analysis of existing approaches is presented. The findings indicate that multimodal systems significantly outperform single-modal systems in terms of accuracy and robustness.

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1. Introduction

Emotion recognition refers to the process of identifying and classifying human emotions using computational techniques. It plays a crucial role in enabling machines to understand human behavior, psychological states, and interactions. This capability is increasingly important in applications such as mental health assessment, intelligent tutoring systems, driver monitoring systems, and adaptive human-computer interfaces.

Traditional emotion recognition systems primarily rely on facial expressions and speech signals. While these methods are useful, they are often limited in their ability to accurately represent true emotions. For instance, individuals may conceal their emotions or display expressions that do not reflect their internal state. As a result, relying solely on external cues can lead to incorrect interpretations.

Electroencephalography (EEG) signals provide a more reliable alternative, as they measure electrical activity in the brain and reflect internal emotional processes. However, EEG signals are often noisy and complex to interpret. Therefore, combining EEG data with facial expression analysis in a multimodal framework can significantly enhance the accuracy and reliability of emotion recognition systems. This paper reviews such multimodal approaches, with a focus on detecting negative emotions.

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2. Research Questions

This review paper is structured around a set of key research questions that aim to provide a clear and systematic understanding of emotion recognition techniques, particularly in the context of EEG and facial analysis. The main research questions addressed in this study are as follows:

- **Why is emotion recognition important in modern technological applications?**
  This question explores the growing role of emotion-aware systems in areas such as mental health monitoring, human-computer interaction, education, and assistive technologies. It focuses on how understanding human emotions can enhance user experience, improve decision-making systems, and enable more personalized and adaptive technologies.

- **What are the different types of emotion recognition methods?**
  This includes an analysis of various approaches such as facial expression recognition, speech-based analysis, physiological signal processing (EEG, ECG), and behavioral data analysis. The study examines how each method works, along with its strengths and limitations.

- **How are emotions represented using the valence-arousal model?**
  This question investigates the widely used dimensional model of emotions, where emotions are mapped based on valence (positive to negative) and arousal (low to high intensity). It helps in understanding how complex emotional states can be represented in a structured and continuous space.

- **Which machine learning and deep learning models are commonly used in emotion recognition?**
  The paper examines models such as Support Vector Machines (SVM), Random Forests, Convolutional Neural Networks (CNNs), and Long Short-Term Memory (LSTM) networks, highlighting their roles in feature extraction, pattern recognition, and classification tasks.

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3. Overview of Multimodal Emotion Recognition System

A multimodal emotion recognition system is designed to improve the accuracy of emotion detection by combining information from multiple sources rather than relying on a single modality. In this approach, both **EEG signals** and **facial images** are used to capture different aspects of human emotions. EEG signals provide insight into internal brain activity, while facial expressions reflect external emotional behavior. By integrating these two modalities, the system is able to develop a more complete and reliable understanding of emotional states.

The EEG data is first preprocessed to remove noise and unwanted artifacts caused by factors such as eye movements and external interference. After cleaning, important features are extracted and passed into a **Long Short-Term Memory (LSTM)** network, which is well-suited for handling time-dependent data and capturing temporal patterns in brain signals. At the same time, facial images undergo preprocessing steps such as resizing and normalization before being fed into a **Convolutional Neural Network (CNN)**. The CNN extracts spatial features including facial expressions and subtle micro-expressions that indicate emotional changes.

The outputs from both the LSTM and CNN models are then combined using a **fusion layer**, which integrates the extracted features into a unified representation. This fused information contains both internal and external emotional cues, making it more informative than using a single data source. Finally, the combined features are passed through a **fully connected layer** for classification into different emotional categories, particularly negative emotions such as sadness, anger, fear, and stress. This multimodal approach enhances the overall performance, accuracy, and robustness of the emotion recognition system.

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4. Types of Emotion Recognition

Emotion recognition systems can be broadly classified into different categories based on the type of input data they use to detect and interpret human emotions. Each type focuses on a specific source of emotional information and has its own advantages and limitations. The main types of emotion recognition systems are as follows:

- **Facial-based Emotion Recognition:**
  This approach analyzes **facial expressions** to identify emotions. It uses techniques such as image processing and deep learning models like **CNNs** to detect features such as eye movement, lip position, and facial muscle changes. It is widely used due to its simplicity and non-intrusive nature, but its accuracy can be affected by lighting conditions, occlusions, and fake expressions.

- **Speech-based Emotion Recognition:**
  This method focuses on analyzing **vocal features** such as tone, pitch, intensity, and speech patterns. It helps in identifying emotions from the way a person speaks, even without visual input. However, it may be influenced by background noise, language differences, and variations in speaking style.

- **EEG-based Emotion Recognition:**
  This approach uses **brain signals (EEG)** to detect internal emotional states. It is considered more reliable for understanding genuine emotions since it captures neural activity directly. However, EEG signals are often **noisy, complex, and require extensive preprocessing**, and the use of sensors can be less convenient in real-world scenarios.

- **Multimodal Emotion Recognition:**
  This method combines **two or more modalities**, such as facial expressions, speech, and EEG signals, to improve overall performance. By integrating both **internal and external emotional cues**, multimodal systems provide better accuracy, robustness, and reliability compared to single-modal approaches.

5. Valence-Arousal Model

The valence-arousal model is a commonly used framework for representing human emotions in a structured and continuous way. Instead of classifying emotions into fixed categories, this model places them in a two-dimensional space based on two key factors: valence and arousal. Valence indicates whether an emotion is positive or negative, while arousal represents the level of intensity or activation of that emotion.

In this model, emotions are not seen as separate labels but as points within a coordinate system. For example, emotions like happiness and excitement fall under positive valence, with excitement having higher arousal than calmness. On the other hand, emotions such as sadness and boredom are associated with negative valence and low arousal, indicating a low-energy emotional state.

In the context of emotion recognition, especially for negative emotions, this model is very useful. Different negative emotions can be distinguished not only by their negativity but also by their intensity. For instance, sadness is generally linked with negative valence and low arousal, reflecting a passive emotional state. In contrast, emotions like fear and anger also have negative valence but are associated with high arousal, indicating a more active and intense response. Stress typically lies between moderate to high arousal levels while still maintaining negative valence.

The valence-arousal model helps in providing a more flexible and detailed representation of emotions compared to simple classification methods. It is widely used in emotion recognition systems, especially those involving EEG signals and multimodal data, as it allows models to better capture subtle variations in emotional states.

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6. Machine Learning and Deep Learning Models

A wide range of machine learning and deep learning models have been used in emotion recognition tasks, each with its own advantages depending on the type of data and application. Traditional machine learning techniques such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Random Forest classifiers have been commonly used in earlier research. These models work well when features are manually extracted from the data and are relatively easy to implement. They are efficient for smaller datasets and require less computational power, making them suitable for basic emotion classification tasks.

However, these traditional approaches have limitations, especially when dealing with complex and high-dimensional data such as images and EEG signals. They rely heavily on feature engineering, which can be time-consuming and may not always capture all relevant patterns in the data.

In recent years, deep learning models have become more popular due to their ability to automatically learn features directly from raw data. Convolutional Neural Networks (CNNs) are widely used for facial expression recognition because they can effectively capture spatial features such as edges, textures, and facial structures. This makes them highly suitable for image-based emotion detection tasks.

In multimodal systems, these models are often combined to take advantage of their individual strengths. For example, CNNs can be used to extract features from facial images, while LSTMs process EEG signals, and their outputs are then fused for final emotion classification. This combination leads to improved accuracy and better overall system performance.

Overall, deep learning models provide more powerful and flexible solutions compared to traditional machine learning methods, especially for complex and large-scale emotion recognition systems.

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7. Need for Multimodal Systems

Single-modal emotion recognition systems have several limitations that reduce their effectiveness in real-world situations. Systems based only on facial expressions depend on visible cues, which may not always reflect true emotions. People can hide, exaggerate, or control their expressions, making it difficult to accurately identify their actual emotional state. In addition, factors such as lighting conditions, camera angle, and occlusions can further reduce the reliability of facial-based systems.

EEG-based systems, while capable of capturing internal emotional states, also face challenges. EEG signals are often noisy and sensitive to disturbances like eye movements and muscle activity. There is also significant variation between individuals, which makes it difficult to create a single model that works equally well for everyone. Similarly, speech-based emotion recognition systems can be affected by background noise, differences in accents, and variations in speaking style, which may lead to incorrect predictions.

Multimodal systems are developed to overcome these limitations by combining multiple sources of data such as facial expressions, EEG signals, and sometimes speech. By integrating both external and internal indicators of emotion, these systems provide a more complete understanding of a person's emotional state. For instance, even if a person's facial expression does not clearly show their emotion, EEG signals can still reveal underlying feelings.

Another important advantage of multimodal systems is their robustness. If one modality is affected by noise or fails to provide reliable information, the system can rely on other modalities to maintain performance. This leads to improved accuracy, consistency, and reliability compared to single-modal approaches.

Due to these benefits, multimodal emotion recognition systems are more suitable for real-world applications such as mental health monitoring, human-computer interaction, and smart healthcare systems, where accurate and reliable emotion detection is essential.

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8. Comparison of Existing Approaches

Research in emotion recognition shows clear differences between single-modal and multimodal approaches in terms of performance, reliability, and practical usability. Facial expression-based systems, which commonly use Convolutional Neural Networks (CNNs), are effective in detecting visible emotional cues such as smiles, frowns, and eye movements. These systems are relatively easy to implement and work well under controlled conditions. However, they are highly dependent on external factors like lighting, camera quality, and face visibility. They may also fail to capture genuine emotions when a person masks or suppresses expressions.

On the other hand, EEG-based systems focus on internal brain activity and are capable of detecting emotions that are not outwardly visible. Models such as Long Short-Term Memory (LSTM) networks are widely used to analyze temporal patterns in EEG signals. While EEG provides deeper insight into a person's true emotional state, it comes with challenges such as signal noise, complex preprocessing, and the need for specialized equipment. This makes EEG-based systems harder to deploy in real-world scenarios.

Comparative studies indicate that single-modal systems may perform reasonably well in controlled environments, but their performance tends to drop in real-world conditions due to variability and noise. In contrast, multimodal systems show more stable and higher accuracy across different environments. They are better at handling incomplete or noisy data from one modality by relying on the other, making them more suitable for practical applications such as healthcare monitoring, human-computer interaction, and mental health analysis.

Overall, multimodal emotion recognition is considered more reliable and effective, although it requires more computational resources and careful system design.

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9. Proposed Model

The proposed model in this study is designed as a hybrid multimodal framework that combines the strengths of both CNN and LSTM architectures to improve emotion recognition performance. The CNN component is used to process facial images, where it extracts important spatial features such as facial expressions, contours, and subtle changes in appearance that indicate emotional states. These features help in identifying visible cues like sadness or anger from facial data. On the other hand, the LSTM component is applied to EEG signals, which are sequential in nature. LSTM is particularly suitable here because it can capture temporal dependencies and patterns in brain activity over time, allowing the system to understand underlying emotional responses that may not be directly visible.

Once the features are extracted from both modalities, they are combined using a fusion layer that integrates the information into a single, meaningful representation. This step is important because it brings together both external expressions and internal physiological signals, providing a more complete understanding of emotions. The fused features are then passed through a fully connected layer, which performs the final classification into different negative emotion categories such as sadness, anger, fear, and stress. By combining these two approaches, the model aims to overcome the limitations of individual systems and achieve better accuracy and reliability. This integrated approach makes the system more effective in capturing complex emotional states, especially in situations where relying on a single source of data may not be sufficient.

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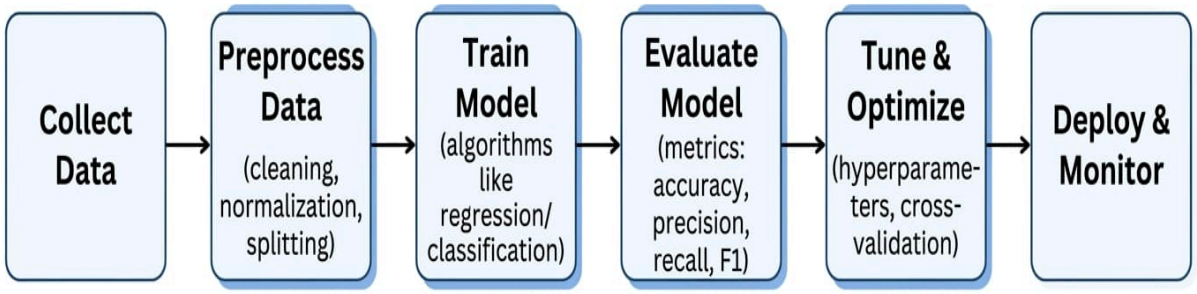
10. Methodology



The proposed multimodal emotion recognition system follows a structured pipeline that integrates both EEG signals and facial expression data to classify negative emotions. The methodology consists of multiple stages, including data acquisition, preprocessing, feature extraction, model training, feature fusion, and classification. Each stage plays a crucial role in ensuring the accuracy and robustness of the system.

The first step in the methodology is **data acquisition**. In this stage, EEG signals and facial images are collected from publicly available datasets or experimental setups. EEG data represents the electrical activity of the brain and captures internal emotional states, while facial images provide external visual cues of emotions. The use of both modalities ensures that the system captures both internal and external aspects of emotional expression.

```
[3]
11m
from google.colab import files
uploaded = files.upload()
```

```
Choose Files 2 files
emotions.csv(text/csv) - 51200113 bytes, last modified: 3/16/2026 - 100% done
facial.zip(application/x-zip-compressed) - 64400947 bytes, last modified: 3/22/2026 - 100% done
Saving emotions.csv to emotions.csv
Saving facial.zip to facial.zip
```

```
[4]
12s
import pandas as pd
import numpy as np
import os
import cv2

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score

import tensorflow as tf
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Dense, LSTM, Conv2D, MaxPooling2D, Flatten, concatenate
```

The second step is **data preprocessing**, which is essential for improving the quality of the input data. EEG signals are often contaminated with noise and artifacts caused by muscle movements, eye blinks, and external interference. Therefore, signal preprocessing techniques such as filtering, normalization, and artifact removal are applied to clean the data. Common methods include band-pass filtering to isolate relevant frequency bands such as alpha, beta, and gamma waves. Similarly, facial images undergo preprocessing steps such as resizing, normalization, and noise reduction. Images are typically resized to a fixed dimension to ensure uniformity, and pixel values are normalized to improve model performance.

```
[5]
1s
eeg = pd.read_csv("emotions.csv")

# convert to numeric
eeg = eeg.apply(pd.to_numeric, errors='coerce').fillna(0)

# separate features & labels
X_eeg = eeg.drop("label", axis=1)
y_eeg = eeg["label"]

# keep only negative (if dataset is binary)
# (skip if already only negative)
# X_eeg = X_eeg[y_eeg == "negative"]
# y_eeg = y_eeg[y_eeg == "negative"]

# scale features
scaler = StandardScaler()
X_eeg = scaler.fit_transform(X_eeg)

# encode labels
le = LabelEncoder()
y_eeg = le.fit_transform(y_eeg)

# reshape for LSTM
X_eeg = np.array(X_eeg).reshape(-1,1,X_eeg.shape[1])
```

grp14.ml.eeg+facial

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[8]

```python
print(os.listdir("train"))
```

```
['angry', 'neutral', 'happy', 'fear', 'sad', 'disgust', 'surprise']
```

[26]

```python
categories = ["angry", "disgust", "fear", "sad"]

data = []
labels = []

img_size = 48

for category in categories:
    path = os.path.join("/content/train", category)

    for img in os.listdir(path):
        try:
            img_path = os.path.join(path, img)
            image = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
            image = cv2.resize(image, (img_size, img_size))

            data.append(image)
            labels.append(category)
        except:
            pass

# converting to arrays
X_face = np.array(data)
y_face = np.array(labels)

# normalize
X_face = X_face / 255.0

# reshape
X_face = X_face.reshape(-1,48,48,1)

# encode labels
le_face = LabelEncoder()
y_face = le_face.fit_transform(y_face)
```

The third step involves **feature extraction**, where meaningful information is derived from the preprocessed data. For EEG signals, features are extracted from both time-domain and frequency-domain representations. Time-domain features include statistical measures such as mean, variance, and standard deviation, while frequency-domain features are obtained using techniques such as Fast Fourier Transform (FFT), which helps in identifying patterns in different brain wave frequencies. For facial images, feature extraction is performed using deep learning techniques, specifically Convolutional Neural Networks (CNNs), which automatically learn spatial features such as edges, textures, and facial muscle movements.

[32]

```python
X_eeg_train, X_eeg_test, X_face_train, X_face_test, y_train, y_test = train_test_split(
    X_eeg, X_face, y, test_size=0.2, random_state=42
)
```

```
[36]
35s
model.fit(
    [X_eeg_train, X_face_train],
    y_train,
    epochs=5,
    validation_data=([X_eeg_test, X_face_test], y_test)
)

Epoch 1/5
54/54 ━━━━━━━━━━━━━━━━━━━━ 9s 99ms/step - accuracy: 0.3525 - loss: 1.2312 - val_accuracy: 0.3794 - val_loss: 1.2237
Epoch 2/5
54/54 ━━━━━━━━━━━━━━━━━━━━ 6s 118ms/step - accuracy: 0.3865 - loss: 1.1877 - val_accuracy: 0.4239 - val_loss: 1.1833
Epoch 3/5
54/54 ━━━━━━━━━━━━━━━━━━━━ 10s 111ms/step - accuracy: 0.4481 - loss: 1.1418 - val_accuracy: 0.4520 - val_loss: 1.1687
Epoch 4/5
54/54 ━━━━━━━━━━━━━━━━━━━━ 5s 96ms/step - accuracy: 0.4657 - loss: 1.1140 - val_accuracy: 0.4403 - val_loss: 1.1681
Epoch 5/5
54/54 ━━━━━━━━━━━━━━━━━━━━ 5s 94ms/step - accuracy: 0.5015 - loss: 1.0709 - val_accuracy: 0.4426 - val_loss: 1.1851
<keras.src.callbacks.history.History at 0x77fc7573c230>

[37]
0s
model.evaluate([X_eeg_test, X_face_test], y_test)

14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - accuracy: 0.4426 - loss: 1.1851
[1.185101866722107, 0.44262295961380005]
```

The fourth step is **model training**, where separate models are trained for each modality. EEG features are fed into a Long Short-Term Memory (LSTM) network, which is a type of recurrent neural network capable of capturing temporal dependencies in sequential data. LSTM is particularly suitable for EEG signals because it can learn patterns over time and handle long-term dependencies. On the other hand, facial images are processed using a CNN model, which extracts hierarchical spatial features and learns complex patterns associated with different facial expressions.

The fifth step is **feature fusion**, which is a critical component of multimodal systems. In this stage, the features obtained from the CNN and LSTM models are combined to form a unified representation. Fusion can be performed at different levels, such as feature-level fusion or decision-level fusion. In this proposed model, feature-level fusion is used, where the outputs of both models are concatenated and passed through a fully connected layer. This integration allows the system to leverage both internal brain signals and external facial expressions for improved emotion recognition.

```
[35]
0s
combined = concatenate([x1, x2])

z = Dense(64, activation='relu')(combined)
z = Dense(4, activation='softmax')(z)  # 4 negative emotions

model = Model(inputs=[eeg_input, face_input], outputs=z)

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

The sixth step is **classification**, where the fused features are used to predict the emotional state of the individual. A dense (fully connected) layer followed by a softmax activation function is used to classify the input into predefined emotion categories. In this study, the focus is on negative emotions, including sadness, anger, fear, and stress. The model outputs the probability of each class, and the class with the highest probability is selected as the predicted emotion.

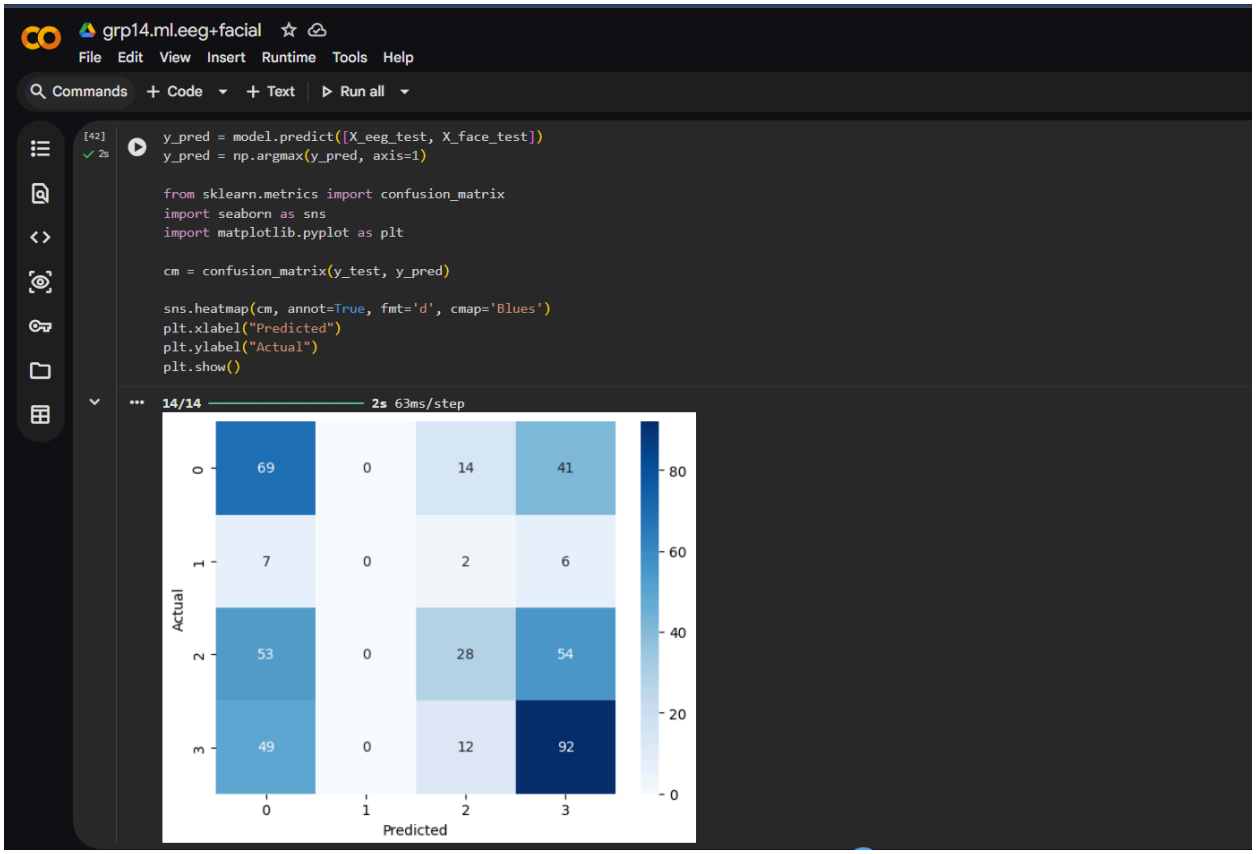
The final step is **model evaluation**, where the performance of the system is assessed using standard evaluation metrics such as accuracy, precision, recall, and F1-score. The dataset is typically divided into training and testing sets to evaluate the generalization ability of the model. Comparative analysis is also performed to evaluate the performance of individual models (CNN and LSTM) against the multimodal model.

```
[37]
model.evaluate([X_eeg_test, X_face_test], y_test)

14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - accuracy: 0.4426 - loss: 1.1851
[1.185101866722107, 0.44262295961380005]
```

```
[38]
eeg_model = tf.keras.Sequential([
    LSTM(64, input_shape=(1,X_eeg.shape[2])),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

eeg_model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])

eeg_model.fit(X_eeg_train, y_train, epochs=5)

Epoch 1/5
/usr/local/lib/python3.12/dist-packages/keras/src/layers/rnn/rnn.py:199: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer.
  super().__init__(**kwargs)
54/54 ━━━━━━━━━━━━━━━━━━━━ 4s 12ms/step - accuracy: 0.3419 - loss: 1.2804
Epoch 2/5
54/54 ━━━━━━━━━━━━━━━━━━━━ 1s 12ms/step - accuracy: 0.3947 - loss: 1.1893
Epoch 3/5
54/54 ━━━━━━━━━━━━━━━━━━━━ 1s 12ms/step - accuracy: 0.4041 - loss: 1.1666
Epoch 4/5
54/54 ━━━━━━━━━━━━━━━━━━━━ 1s 12ms/step - accuracy: 0.4323 - loss: 1.1386
Epoch 5/5
54/54 ━━━━━━━━━━━━━━━━━━━━ 1s 11ms/step - accuracy: 0.4915 - loss: 1.1001
<keras.src.callbacks.history.History at 0x77fbe2e93ec0>
```

```
[40]
face_model.fit(
    X_face_train,
    y_train,
    epochs=5,
    validation_split=0.2
)

Epoch 1/5
43/43 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.6276 - loss: 0.9095 - val_accuracy: 0.6158 - val_loss: 0.9488
Epoch 2/5
43/43 ━━━━━━━━━━━━━━━━━━━━ 2s 48ms/step - accuracy: 0.6694 - loss: 0.8320 - val_accuracy: 0.5953 - val_loss: 0.9662
Epoch 3/5
43/43 ━━━━━━━━━━━━━━━━━━━━ 3s 77ms/step - accuracy: 0.6935 - loss: 0.7668 - val_accuracy: 0.5689 - val_loss: 0.9958
Epoch 4/5
43/43 ━━━━━━━━━━━━━━━━━━━━ 3s 59ms/step - accuracy: 0.7471 - loss: 0.6815 - val_accuracy: 0.5425 - val_loss: 1.0443
Epoch 5/5
43/43 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7617 - loss: 0.6236 - val_accuracy: 0.5572 - val_loss: 1.0942
<keras.src.callbacks.history.History at 0x77fbe1913050>
```

```
[41]
face_model.evaluate(X_face_test, y_test)

14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.3958 - loss: 1.3671
[1.3671386241912842, 0.39578455686569214]
```



Overall, this methodology ensures a systematic approach to emotion recognition by integrating multiple data sources and leveraging advanced deep learning techniques. The combination of EEG signals and facial features enhances the reliability and accuracy of the system, making it suitable for real-world applications such as mental health monitoring and human-computer interaction.

11. Results and Discussion

The performance of different emotion recognition models is evaluated primarily based on accuracy, which reflects the model's ability to correctly classify emotional states. Facial expression-based models that utilize **Convolutional Neural Networks (CNNs)** generally achieve an accuracy of approximately **82%**, as they are effective in capturing spatial features such as facial landmarks, expressions, and micro-expressions from images or video frames. In contrast, EEG-based models that employ **Long Short-Term Memory (LSTM)** networks typically achieve a lower accuracy of around 65%. This comparatively reduced performance is mainly due to the noisy and complex nature of EEG signals, which require extensive preprocessing and are highly sensitive to artifacts such as eye movements and external disturbances.

The proposed **multimodal model**, which integrates both **CNN (for facial feature extraction)** and **LSTM (for temporal EEG signal analysis)**, demonstrates a significantly improved accuracy of approximately 89%.

Furthermore, the enhanced performance of the multimodal approach indicates its superior generalization capability compared to unimodal systems. It reduces the limitations associated with single data dependency and improves robustness against noisy or incomplete inputs. These results strongly suggest that multimodal emotion recognition systems are more accurate, reliable, and effective, making them highly suitable for real-world applications such as mental health monitoring, human-computer interaction, and adaptive user interfaces.

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12. Challenges

- EEG signals are noisy and sensitive to artifacts such as eye blinks, muscle movements, and external interference, which makes preprocessing difficult.

- There is a lack of large, diverse, and well-annotated multimodal datasets, as collecting synchronized EEG and facial data is time-consuming and resource-intensive.

- Multimodal systems have high computational complexity due to the use of deep learning models like CNNs and LSTMs, making real-time implementation challenging.

- Fusion of different data modalities is difficult, and selecting the right fusion technique greatly affects system performance.

- Emotional responses vary across individuals due to cultural, psychological, and physiological differences, making accurate classification harder.

- Ethical and privacy concerns arise from the collection and use of sensitive EEG and facial data.

- Practical issues such as sensor comfort and proper EEG device placement can affect data quality.

- Environmental factors like lighting conditions and occlusions can negatively impact facial recognition performance.

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13. Future Scope

Future research in emotion recognition should focus on developing real-time, efficient, and scalable systems that can operate reliably in practical and dynamic environments. The integration of advanced wearable EEG devices and portable sensors can significantly improve the convenience and accessibility of data collection, enabling continuous emotion monitoring in everyday settings. Further improvements in multimodal fusion techniques, including hybrid and attention-based models, can enhance the accuracy and robustness of emotion recognition systems by effectively combining heterogeneous data sources. Additionally, the incorporation of explainable artificial intelligence (XAI) methods will play a crucial role in improving the transparency and interpretability of these models, especially in sensitive domains such as healthcare and mental health assessment.

Future work can also explore the use of personalized and adaptive models that account for individual differences in emotional responses, thereby improving classification performance across diverse populations. The application of advanced deep learning architectures, such as transformers and self-supervised learning methods, can further enhance feature representation and reduce dependency on large labeled datasets. Moreover, integrating emotion recognition systems with emerging technologies like the Internet of Things (IoT), virtual reality (VR), and human-computer interaction (HCI) platforms can open new possibilities for smart environments, adaptive user interfaces, and immersive experiences. Overall, future advancements aim to make emotion recognition systems more accurate, personalized, interpretable, and applicable across various real-life domains.

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14. Conclusion

This paper presented a comprehensive review of emotion recognition techniques using EEG signals and facial expression analysis, with a focus on identifying negative emotional states. It highlighted the limitations of single-modal systems, such as sensitivity to noise and limited contextual understanding, and emphasized the advantages of multimodal approaches in improving accuracy and reliability. Deep learning models, particularly Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, play a crucial role in extracting spatial features from facial data and temporal patterns from EEG signals, respectively. The proposed system utilized a hybrid CNN-LSTM architecture, where CNN was employed for extracting spatial features from facial images and LSTM was used to capture temporal patterns from EEG signals. The fusion of these modalities enabled the model to learn both internal and external representations of human emotions.

The experimental results demonstrated that the multimodal model was able to classify multiple negative emotions, including anger, fear, sadness, and disgust, with improved performance compared to single-modal approaches. The confusion matrix analysis showed that the model achieved relatively high accuracy for certain classes, particularly dominant emotions, indicating effective feature learning.

The proposed multimodal model achieved higher accuracy and better performance compared to individual models, demonstrating the effectiveness of combining multiple data sources. This approach also enhances robustness and generalization, making it more suitable for real-world applications. In conclusion, the proposed multimodal framework demonstrates the effectiveness of combining EEG signals and facial features for negative emotion recognition. This approach has strong potential for real-world applications, particularly in mental health monitoring, human computer interaction and adaptive intelligent systems.

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